

Cylindrical Decomposition of a Tilted Finite Gaussian Beam

1 Introduction

A practical laser beam has a localized transverse profile and is commonly approximated by a Gaussian field distribution. When such a beam is launched at an angle relative to the axis of a cylindrical optical system, such as an optical fiber, its phase varies across the transverse input plane. Expressing this tilted Gaussian field in cylindrical coordinates reveals that it can be decomposed into a sum of Gaussian-modulated cylindrical Bessel components.

This decomposition follows from the Jacobi–Anger expansion, which expresses a tilted plane-wave phase factor as a series of cylindrical harmonics. Each term in the series contains a Bessel radial factor, a Gaussian envelope, and an azimuthal phase factor. The resulting components have the mathematical form

$$J_m(k_{\perp}\rho) \exp\left(-\frac{\rho^2}{w_0^2}\right) \exp(jm\phi). \quad (1)$$

This paper develops the derivation from the incident finite Gaussian beam to its cylindrical Bessel decomposition. It also explains the physical meaning of each factor and clarifies the distinction between a mathematical cylindrical decomposition and the excitation of guided fiber modes.

2 The Gaussian Function

The Gaussian function describes a bell-shaped profile with a maximum at the center and a gradual decay with distance from the center, as illustrated in Fig. 1.

At the beam waist, the transverse electric-field amplitude of a fundamental Gaussian beam can be written in Cartesian coordinates as

$$E(x, y) = E_0 \exp\left[-\frac{x^2 + y^2}{w_0^2}\right], \quad (2)$$

where E_0 is the electric-field amplitude at the beam center, w_0 is the beam-waist radius, and x and y are the transverse coordinates. Using the cylindrical-coordinate relation

$$\rho^2 = x^2 + y^2, \quad (3)$$

the Gaussian field at the beam waist becomes

$$E(\rho) = E_0 \exp\left(-\frac{\rho^2}{w_0^2}\right). \quad (4)$$

More generally, the radial envelope of a propagating Gaussian beam may be expressed as

$$E(\rho, z) = A(z) \exp\left[-\frac{\rho^2}{w^2(z)}\right], \quad (5)$$

where $A(z)$ is the on-axis field amplitude at $\rho = 0$, and $w(z)$ is the beam radius at axial position z . A larger $w(z)$ produces a broader field distribution, whereas a smaller $w(z)$ produces a more tightly confined field. In Eq. (5), $A(z)$ may include the longitudinal phase, wavefront-curvature phase, and Gouy phase when the complete Gaussian-beam field is required.

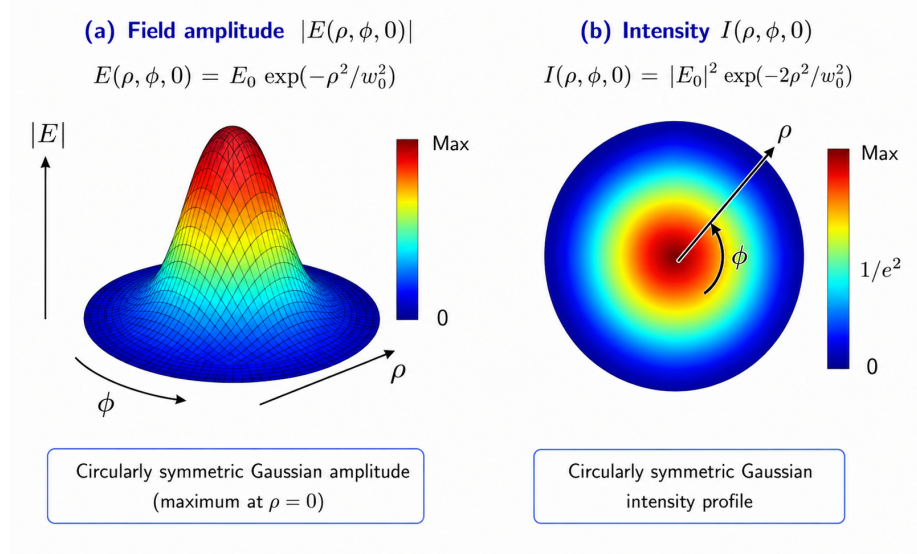


Figure 1: Gaussian beam amplitude and intensity profiles.

As illustrated in Fig. 2, the Gaussian envelope provides smooth radial confinement. Although the mathematical Gaussian profile extends to infinity, it decays rapidly with radial distance and has finite total power.

Let a laser beam be incident on the entrance facet of an optical fiber. Let the fiber axis define the z -direction, and let the input facet be located at $z = 0$. The choice $z = 0$ is simply a coordinate convention.

At the beam waist, the corresponding intensity distribution is proportional to the squared magnitude of the electric field:

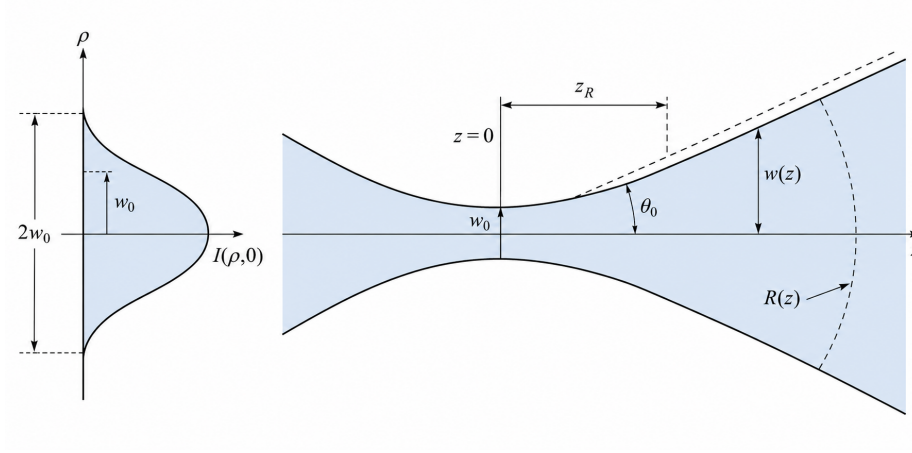


Figure 2: Gaussian beam propagation and characteristic parameters.

$$I_G(\rho) \propto |E_G(\rho)|^2 = |E_0|^2 \exp\left(-\frac{2\rho^2}{w_0^2}\right). \quad (6)$$

The incident tilted Gaussian field can be represented as

$$E_{\text{in}}(\rho, \phi, z) = E_0 \exp\left(-\frac{\rho^2}{w_0^2}\right) \exp(j\mathbf{k} \cdot \mathbf{r}), \quad (7)$$

where $\mathbf{k}$ is the wave vector and $k = |\mathbf{k}|$ is its magnitude in the relevant medium. The wave vector can be separated into transverse and longitudinal components:

$$\mathbf{k} = \mathbf{k}_\perp + k_z \hat{\mathbf{z}}. \quad (8)$$

The transverse wave number $k_\perp = |\mathbf{k}_\perp|$ and the longitudinal wave number k_z satisfy

$$k_\perp = k \sin \theta_{\text{in}}, \quad (9)$$

and

$$k_z = k \cos \theta_{\text{in}}, \quad (10)$$

where θ_{in} is measured relative to the fiber axis. At the input plane $z = 0$, the longitudinal phase equals unity, leaving only the transverse phase factor

$$\exp(j\mathbf{k}_\perp \cdot \mathbf{r}_\perp). \quad (11)$$

This factor represents the linear phase variation across the input facet caused by the angular launch. Therefore, Eq. (7) becomes

$$E_{\text{in}}(\rho, \phi, 0) = E_0 \exp\left(-\frac{\rho^2}{w_0^2}\right) \exp(j\mathbf{k}_{\perp} \cdot \mathbf{r}_{\perp}). \quad (12)$$

3 Finite Gaussian Beam at Normal Incidence

Let the laser beam be normally incident on the input plane, such that $\theta_{\text{in}} = 0^\circ$. As illustrated in Fig. 3, the transverse wave-vector component vanishes, $k_{\perp} = 0$, and the incident wave vector is entirely longitudinal, $k_z = k$. Consequently, $\exp(j\mathbf{k}_{\perp} \cdot \mathbf{r}_{\perp}) = 1$, and Eq. (12) reduces to

$$E_{\text{in}}(\rho, \phi, 0) = E_0 \exp\left(-\frac{\rho^2}{w_0^2}\right). \quad (13)$$

The field amplitude and intensity decrease continuously with increasing radial distance. The field is therefore transversely localized and has finite total power, although its mathematical profile does not terminate at a finite value of ρ .

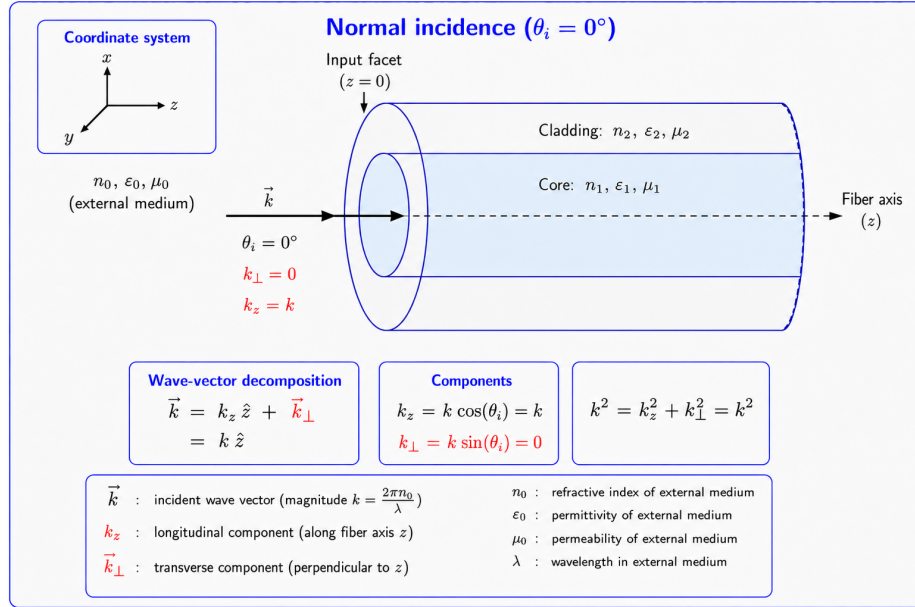


Figure 3: Wave-vector representation of an optical beam launched normally into the core of a step-index optical fiber.

4 Finite Gaussian Beam at Oblique Incidence

Suppose that the beam is launched at an angle θ_{in} relative to the fiber axis, as illustrated in Fig. 4.

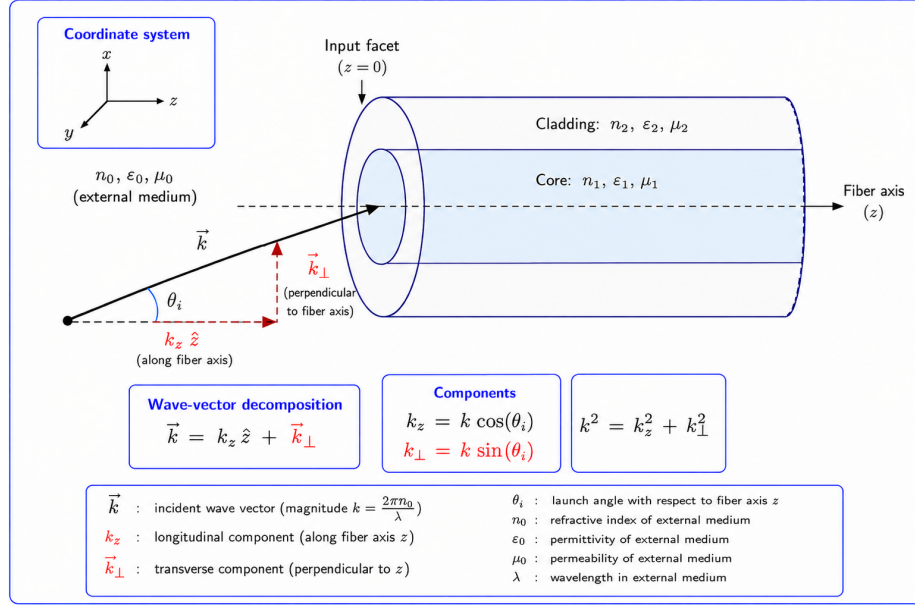


Figure 4: Wave-vector decomposition of an optical beam incident obliquely on the input facet of a step-index optical fiber.

For sufficiently small launch angles expressed in radians,

$$k_\perp \approx k\theta_{\text{in}}. \quad (14)$$

4.1 Transverse Phase in Cartesian Coordinates

Let the transverse wave vector point in the azimuthal direction ϕ_0 . It can be written as

$$\mathbf{k}_\perp = k_\perp (\cos \phi_0 \hat{\mathbf{x}} + \sin \phi_0 \hat{\mathbf{y}}). \quad (15)$$

The transverse position vector is

$$\mathbf{r}_\perp = x\hat{\mathbf{x}} + y\hat{\mathbf{y}}. \quad (16)$$

Their dot product is

$$\mathbf{k}_\perp \cdot \mathbf{r}_\perp = k_\perp (x \cos \phi_0 + y \sin \phi_0). \quad (17)$$

4.2 Transformation to Cylindrical Coordinates

Using

$$x = \rho \cos \phi, \quad y = \rho \sin \phi, \quad (18)$$

the transverse coordinate factor becomes

$$x \cos \phi_0 + y \sin \phi_0 = \rho (\cos \phi \cos \phi_0 + \sin \phi \sin \phi_0). \quad (19)$$

Using the trigonometric identity

$$\cos(\phi - \phi_0) = \cos \phi \cos \phi_0 + \sin \phi \sin \phi_0, \quad (20)$$

we obtain

$$\mathbf{k}_\perp \cdot \mathbf{r}_\perp = k_\perp \rho \cos(\phi - \phi_0). \quad (21)$$

Substituting Eq. (21) into Eq. (12), the tilted finite Gaussian beam at the input plane can be written in cylindrical coordinates as

$$E_{\text{in}}(\rho, \phi, 0) = E_0 \exp\left(-\frac{\rho^2}{w_0^2}\right) \exp[jk_\perp \rho \cos(\phi - \phi_0)]. \quad (22)$$

Equation (22) contains two physically distinct factors. The first is the Gaussian amplitude envelope,

$$\underbrace{\exp\left(-\frac{\rho^2}{w_0^2}\right)}_{\text{finite Gaussian envelope}},$$

whereas the second is the transverse phase ramp produced by the angular launch,

$$\underbrace{\exp[jk_\perp \rho \cos(\phi - \phi_0)]}_{\text{phase produced by the angular launch}}.$$

At this stage, the field remains a tilted Gaussian beam. The tilt changes the phase distribution but not the intensity distribution at the input plane, because the angular phase factor has unit magnitude:

$$|\exp[jk_\perp \rho \cos(\phi - \phi_0)]| = 1. \quad (23)$$

4.3 Cylindrical Expansion Using the Jacobi–Anger Identity

The Jacobi–Anger expansion is

$$\exp(jx \cos \alpha) = \sum_{m=-\infty}^{\infty} j^m J_m(x) \exp(jm\alpha), \quad (24)$$

where $J_m(x)$ is the Bessel function of the first kind of integer order m . Setting

$$x = k_{\perp} \rho, \quad (25)$$

and

$$\alpha = \phi - \phi_0, \quad (26)$$

the angular phase factor becomes

$$\exp[jk_{\perp} \rho \cos(\phi - \phi_0)] = \sum_{m=-\infty}^{\infty} j^m J_m(k_{\perp} \rho) \exp[jm(\phi - \phi_0)]. \quad (27)$$

The azimuthal exponential may be separated as

$$\exp[jm(\phi - \phi_0)] = \exp(jm\phi) \exp(-jm\phi_0). \quad (28)$$

Therefore,

$$\exp[jk_{\perp} \rho \cos(\phi - \phi_0)] = \sum_{m=-\infty}^{\infty} j^m \exp(-jm\phi_0) J_m(k_{\perp} \rho) \exp(jm\phi). \quad (29)$$

4.4 Gaussian-Modulated Bessel Components

Substituting Eq. (29) into Eq. (22) gives

$$E_{\text{in}}(\rho, \phi, 0) = E_0 \exp\left(-\frac{\rho^2}{w_0^2}\right) \sum_{m=-\infty}^{\infty} j^m \exp(-jm\phi_0) J_m(k_{\perp} \rho) \exp(jm\phi). \quad (30)$$

Because the Gaussian envelope is independent of m , it may be included inside the summation:

$$E_{\text{in}}(\rho, \phi, 0) = \sum_{m=-\infty}^{\infty} E_0 j^m \exp(-jm\phi_0) J_m(k_{\perp} \rho) \exp\left(-\frac{\rho^2}{w_0^2}\right) \exp(jm\phi). \quad (31)$$

Define the coefficient

$$A_m = E_0 j^m \exp(-jm\phi_0). \quad (32)$$

The incident field then becomes

$$E_{\text{in}}(\rho, \phi, 0) = \sum_{m=-\infty}^{\infty} A_m J_m(k_{\perp} \rho) \exp\left(-\frac{\rho^2}{w_0^2}\right) \exp(jm\phi) \quad (33)$$

Each term in this mathematical decomposition has the form

$$E_m(\rho, \phi, 0) = A_m J_m(k_{\perp} \rho) \exp\left(-\frac{\rho^2}{w_0^2}\right) \exp(jm\phi). \quad (34)$$

The complete tilted Gaussian beam is not a single component of the form given in Eq. (34); rather, it is the coherent sum over all integer values of m . Furthermore, the terms in this Jacobi–Anger expansion are cylindrical harmonic components of the incident field and should not automatically be identified with the guided eigenmodes of the fiber.

4.5 Physical Meaning of the Field Components

Each cylindrical component contains three principal factors.

4.5.1 Bessel Radial Factor

The radial factor is

$$J_m(k_{\perp} \rho). \quad (35)$$

This factor appears because the tilted transverse phase is expressed in a cylindrical basis. The launch angle determines the transverse wave number through

$$k_{\perp} = k \sin \theta_{\text{in}}. \quad (36)$$

A larger launch angle therefore produces a larger transverse spatial frequency, provided that k remains fixed.

For fiber-input analysis, it is useful to introduce the dimensionless transverse parameter

$$\chi = ak_{\perp}, \quad (37)$$

where a is the fiber-core radius. Consequently,

$$\chi = ak \sin \theta_{\text{in}}. \quad (38)$$

If λ_0 is the free-space wavelength, then

$$k_0 = \frac{2\pi}{\lambda_0} \quad (39)$$

is the free-space wave number. In an external launch medium with refractive index n_{ext} , the wave number is

$$k_{\text{ext}} = n_{\text{ext}} k_0 = \frac{2\pi n_{\text{ext}}}{\lambda_0}. \quad (40)$$

When θ_{in} is the angle in that external medium,

$$k_{\perp} = k_{\text{ext}} \sin \theta_{\text{in}}, \quad (41)$$

and hence

$$\chi = a k_{\text{ext}} \sin \theta_{\text{in}}. \quad (42)$$

Equivalently, let λ_{ext} denote the wavelength in the external medium, where $\lambda_{\text{ext}} = \lambda_0/n_{\text{ext}}$. It follows that $k_{\text{ext}} = 2\pi/\lambda_{\text{ext}}$. The dimensional and normalized transverse quantities are related by

$$k_{\perp} = \frac{\chi}{a}. \quad (43)$$

The Bessel factor may therefore be written as

$$J_m(k_{\perp} \rho) = J_m\left(\frac{\chi \rho}{a}\right). \quad (44)$$

This expression is dimensionally consistent because both χ and ρ/a are dimensionless. The individual cylindrical component becomes

$$E_m(\rho, \phi, 0) = A_m J_m\left(\frac{\chi \rho}{a}\right) \exp\left(-\frac{\rho^2}{w_0^2}\right) \exp(jm\phi). \quad (45)$$

4.5.2 Gaussian Envelope

The Gaussian envelope is

$$\exp\left(-\frac{\rho^2}{w_0^2}\right). \quad (46)$$

This factor originates directly from the localized transverse profile of the incident laser beam and limits the effective radial extent of every cylindrical component.

4.5.3 Azimuthal Phase Factor

The azimuthal phase factor is

$$\exp(jm\phi). \quad (47)$$

This factor describes the angular phase variation around the optical axis. Over one complete revolution, $\phi \rightarrow \phi + 2\pi$, the phase changes by

$$\Delta\Phi = 2\pi m. \quad (48)$$

Thus, each separated cylindrical harmonic has an azimuthal phase winding of integer order m . Nevertheless, the original tilted Gaussian beam is the coherent sum of all these harmonics and should not be interpreted as a single OAM eigenstate.

4.6 Longitudinal Propagation

The Jacobi–Anger expansion establishes the field decomposition only at the input plane. If a formal longitudinal propagation factor is assigned to each cylindrical component, the field may be represented as

$$E(\rho, \phi, z) = \sum_{m=-\infty}^{\infty} A_m J_m(k_{\perp}\rho) \exp\left(-\frac{\rho^2}{w_0^2}\right) \exp(jm\phi) \exp(-j\beta_m z). \quad (49)$$

Equation (49) is a formal extension rather than a result of the Jacobi–Anger identity itself. In a fiber, the allowed propagation constants β_m , radial functions, and excitation amplitudes must be determined from the fiber eigenvalue problem and from overlap integrals between the incident field and the guided eigenmodes.

5 Conclusion

A tilted finite Gaussian beam can be decomposed into Gaussian-modulated cylindrical Bessel components by expressing its transverse phase in cylindrical coordinates and applying the Jacobi–Anger expansion. The principal input-plane result is

$$E_{\text{in}}(\rho, \phi, 0) = \sum_{m=-\infty}^{\infty} A_m J_m(k_{\perp}\rho) \exp\left(-\frac{\rho^2}{w_0^2}\right) \exp(jm\phi), \quad (50)$$

where

$$A_m = E_0 j^m \exp(-jm\phi_0). \quad (51)$$

Using the normalized transverse parameter $\chi = ak_{\perp}$, the same result may be written as

$$E_{\text{in}}(\rho, \phi, 0) = \sum_{m=-\infty}^{\infty} A_m J_m \left(\frac{\chi \rho}{a} \right) \exp \left(-\frac{\rho^2}{w_0^2} \right) \exp(jm\phi). \quad (52)$$

The Gaussian envelope is inherited from the finite laser excitation, the Bessel factor results from the cylindrical expansion of the tilted phase, and the azimuthal factor identifies the angular harmonic order. This decomposition provides an analytical representation of the incident field; the amplitudes of the actual guided fiber modes must subsequently be calculated using modal overlap integrals.

Ibrahim Barka
August 2026